



SpaceSuit XR

SpaceSuit View

Introduction and instructions

Private alpha · Meta Quest

View your own Gaussian-splat scenes on a standalone Quest, at room scale, without a PC.

Welcome to SpaceSuit View

SpaceSuit View puts captured places and objects in your room. It runs entirely on the headset: no PC, no cable, no account.

A **Gaussian splat** is a 3D capture made of hundreds of thousands of small, soft, coloured points. Together they reproduce a real object or place with its light and texture, which a regular 3D model often loses. SpaceSuit View shows these scenes in **passthrough**, so you see the scene and your room together. You can walk around it, pick it up, scale it, step inside it and measure it.

What you can do

Explore two sample scenes

A May beetle scan and a potted plant come with the app and work straight away.

Bring your own scenes

Import PLY files or SuperSplat downloads from the headset's Download folder. The app converts them on the headset.

Move, scale and step inside

Grab a scene with one hand, resize it with the thumbstick, see it whole or walk into it.

Measure at real size

Measure between two points, and enter a known distance to rescale the whole scene to real size.

What you need

- A **Meta Quest 3** (recommended). Quest 3S and Quest 2 are also supported in the alpha; see the limitations on page 8.
- **Both controllers.** Hand tracking is not supported.
- A clear play area, and an invitation to the SpaceSuit View alpha sent to your Meta account email.

Your files stay on your headset

SpaceSuit View has no internet access. It does not send your scenes, your settings or your movements anywhere, and it has no accounts, analytics or advertising. It reads and writes scene files only in the headset's **Download** folder, and it deletes a file only when you confirm. **Browse SuperSplat** opens the separate Quest Browser, which follows its own policies.

This is an early alpha. It has been tested on one Meta Quest 3 by its developer. Expect rough edges, and please tell us what goes wrong (page 8).

Install and first launch

From invitation to your first scene in a few minutes.

1. Install

- 1 Accept the alpha invitation sent to your Meta account email.
- 2 In the headset, open the **Library**, find **SpaceSuit View** and install it.

2. Open the app

- 1 Open SpaceSuit View. The SpaceSuit XR logo shows over your room while it loads.
- 2 The **May beetle** sample appears in front of you, with the **Welcome to SpaceSuit View** card. The card shows the controllers and the three things to know first.
- 3 Read it, then select **Got it** at the top right (or press **B** or **Y**). The card shows only once; its contents are always on the menu's **Help** tab.

3. Allow file access

To see your own scenes in the **Download** folder, Android asks for **All files access**. After the welcome card, the menu opens on the **Scenes** tab with an **Allow file access** row.

- 1 Point at **Allow file access** and pull the **trigger**.
- 2 On the Android page that opens, switch on **Allow access to manage all files**.
- 3 Go back to SpaceSuit View. Your Download files now show in the menu.

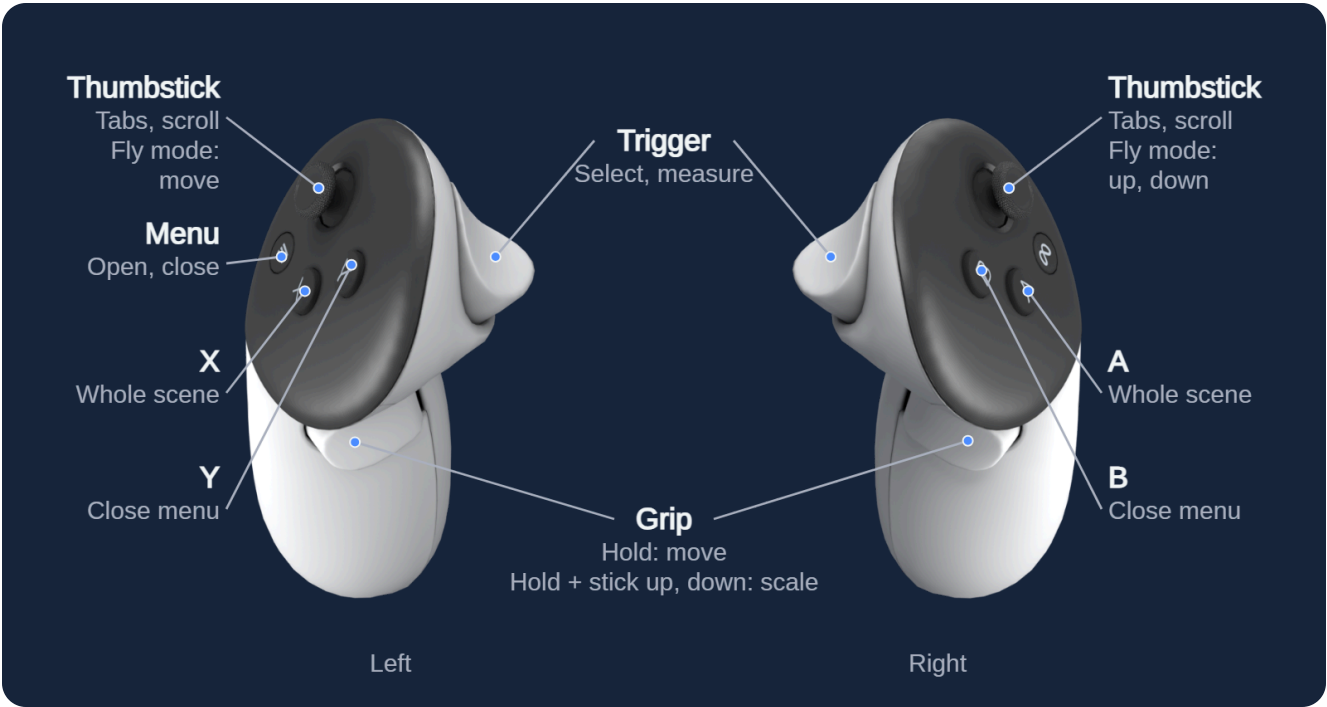
You can skip this step for now: the two sample scenes work without file access. The app asks again each time it starts until access is allowed.

4. Try the samples

Open the menu with the **menu button** on the left controller. The **Scenes** tab lists the two samples, **May beetle** and **Potted plant**, marked *Sample*. Point at one and pull the trigger to load it. Samples can't be deleted.

Your controllers

Point and pull the trigger to use the menu. Grip to grab the scene.



TO	DO THIS
Open or close the menu	Menu button on the left controller. B or Y also closes it.
Use the menu	Point with a controller and pull the trigger . Push a thumbstick left or right to change tabs, up or down to scroll.
Move the scene	Hold a grip and move your hand.
Make it bigger or smaller	While holding a grip, push the thumbstick up or down .
See the whole scene	Press A or X . To come back close: View > Enter scene .
Fly through it	View > Fly mode : the left stick moves you, the right stick goes up and down. Select it again to stop.
Measure	Touch a point with the controller tip and pull the trigger , then a second point.

The scene ignores the controllers while the menu is open, so close the menu to grab or measure.

Five tabs

Open the menu with the left controller's menu button. Push a thumbstick left or right to move between tabs.



Scenes

Everything you can load. At the top, **Recenter** places the loaded scene in front of you and **Unload** removes it from view. Below are the two samples, then your converted scenes from Download. Each row shows the number of splats and the file size; the loaded scene is marked *Loaded*. Select a row to load it.

Import

Turns PLY files and SuperSplat zips into scenes the app can show. It has **Browse SuperSplat**, the PLY and zip files found in Download, and two settings: **Splats to keep** and **Upright fix**. See the next page.

View

- **Grab outline**: shows the scene's box while you grab it.
- **Recenter**: an outside overview, with the whole scene in front of you.
- **Enter scene**: puts the scene's centre 2 m ahead, so you can step inside it.
- **Fly mode**: the left thumbstick moves you, the right thumbstick goes up and down.

About

The app version, **Send feedback**, which opens our feedback page in the Quest Browser, and the credits for the controller drawing and the sample scenes.

Help

The controller drawing and a short reminder of every control. Nothing on this tab is a button.

Deleting files

Delete on a row in Scenes or Import removes that file from the headset. You confirm first (**Keep it** cancels). It can't be undone.

Import a scene

Download a splat, import it once, and it stays on the Scenes tab.

Get a scene from SuperSplat

- 1 Open the menu and go to the **Import** tab. Select **Browse SuperSplat**. The Quest Browser opens at superspl.at.
- 2 Pick a scene and download it as **PLY file**. You get a zip, which lands in Download.
Not the **SOG** / "Source file" download: the app can't read it yet.
- 3 Go back to SpaceSuit View. The zip appears on the Import tab.

You can also copy **.ply** files into the headset's Download folder from a computer.

Import and view it

- 1 On the Import tab, check **Splats to keep: 500k**, the default, runs smoother; **750k** shows more detail.
- 2 Select your file. A progress bar runs; most scenes take a few seconds. **Cancel conversion** stops it. The scene opens when it is done.
- 3 Next time, find it on the **Scenes** tab and select it to load it.

Supported files

FORMAT	SUPPORTED
Standard PLY	Yes
SuperSplat compressed PLY	Yes
SuperSplat "PLY file" zip	Yes
SOG, SPZ, KSPLAT, SPLAT	Not yet

Scene upside down? Scenes from SuperSplat need the **Upright fix**, which is on by default; some other sources load upside down with it on. Turn **Upright fix** off on the Import tab, delete the converted scene on the Scenes tab, and import the file again.

Importing a zip needs free space for a temporary copy of the scene inside it. The copy is removed when the import ends.

Move, explore and measure

Close the menu first: the scene responds to the controllers only while the menu is closed.

Place it

- **Grab:** hold a grip and move your hand. The scene follows. Turn **View > Grab outline** on to see its box while you move it.
- **Scale:** keep holding the grip and push the thumbstick up to enlarge, down to shrink.
- **Recenter:** **Scenes > Recenter** or **View > Recenter** brings the scene back in front of you.

Explore it

- **Walk around it.** The scene stays fixed in your room, so you can lean in and look from any side.
- **See it whole:** press A or X. Large scenes open about 2 m in front of you so you can step inside; A or X shows the whole thing.
- **Step inside:** **View > Enter scene** puts the scene's centre 2 m ahead.
- **Fly:** **View > Fly mode.** The left stick moves you, the right stick goes up and down. Select it again to stop.

Measure and set real size

- 1 Touch a point in the scene with the **controller tip** and pull the **trigger**.
- 2 Touch a second point and pull the trigger again. The distance between them shows.
- 3 If you know the real distance, set it with the **thumbstick**, then pull the trigger to apply. The whole scene rescales to real size.

Each scene starts at the app's default size when it loads. Size, position and measurements are not remembered between scenes or between sessions yet.

Limitations and feedback

What to expect from this alpha, and how to reach us.

Smoothness

Small scenes run at the Quest's full 72 frames per second; large ones do not yet. Measured on a Quest 3 with the samples, each seen from about 0.85 m:

SCENE	FRAMES PER SECOND
Potted plant (269,000 splats)	71 to 73, the full rate
May beetle (500,000 splats)	44 to 47

Scenes imported at 500,000 splats run at about the beetle's rate; 750,000 is slower still. You may notice judder when you turn your head quickly or fly. If that is uncomfortable, stop, and tell us. Making it faster is planned.

Scene size

Every imported scene is reduced to 500,000 splats by default (750,000 optional). Faces and overall shape hold up well, but a very large scene loses some fine texture. Scenes of several million splats shown unreduced are not supported.

Headsets

All figures were measured on a Quest 3. Quest 3S has the same chip but has not been tested yet. On a Quest 2, passthrough is black and white and scenes run slower than the figures above.

Send us feedback

Open the menu, go to **About** and select **Send feedback**. It opens spacesuitxr.com/alpha in the Quest Browser. Tell us the scene, its size, your headset and what you were doing when something went wrong.

Credits

May beetle sample: MAY-BEETLE-HIRES by Huib Schippers, superspl.at, CC BY 4.0, reduced to 500,000 splats.

Potted plant sample: Potted Plant 02 by Rico Cilliers, Poly Haven, CC0.

Controller drawing: controller models © Meta Platform Technologies, LLC.

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